Date Submitted: Not Submitted

Project Title
Pilot Scale Verification of Modified Pretreatment Process for Pineapple Fibers

Project Description

The declines situation of the local textile industry serves as the driving force for the Philippine Textile Research Institute (PTRI) to continuously undertake research and development activities to come up with textile products from natural sources that could help boost the industry.

PTRI has been steadfast in its effort to judiciously utilize natural sources in view of their advantages over synthetic fibers. Not only are they environment friendly which is the world's one of the priority at this time, but more importantly, the utilization of these agricultural wastes would minimize the production and emission of toxic gases while providing additional source of income and livelihood for the farmers in the countryside.

Consistent to its goal, the Institute has explored the potential and established pretreatment technology of the different indigenous fibers such abaca, banana, pineapple, saluyot and other fibers for textile manufacture. However, the established pretreatment technology are still too costly for textile production, Improvement or modification of the existing technology or establishment of simple method could lower the cost of production, and thereby redound to cheaper product but of acceptable quality.

Modified pretreatment technology for pineapple fiber was undertaken in the laboratory scale and shows promising result. Likewise, the utilization of spent liquor was also explored in the laboratory scale and test results show that it can be utilized or recycled for pretreatment pineapple fiber.

It is therefore the objective of this project to verify and assess the commercial applicability of the modified pretreatment technology and the utilization of spent liquor for degumming pineapple fiber. It is also aimed at establishing a competitive and more environment friendly process for pineapple fibers by reducing effluent. Improvement and modification in the pretreatment of pineapple fiber for the processing of textile product to make it cost competitive can very well guarantee a fruitful endeavor for the textile industry.

The project needs equipment/capital outlay of Php 2,000,000.00 for the provision of stainless steel stock tank, chemical pump, piping and fittings and replacement or concreting of the dilapidated wooden platform of the existing degumming vessel. The provision is necessary for the implementation/pretreatment of pineapple fiber using spent liquor as well as safety for the machine operator/researcher.

Project Objective (summary)

General Objective:
To conduct pilot scale verification of the modified pretreatment process for pineapple fiber.

Specific Objectives:
1) To conduct pilot scale verification of the pretreatment of pineapple fibers using spent liquor;
2) To evaluate spinning performance and property of pretreated pineapple-blended fibers.

Project Duration	Jan 01, 2017 to Dec 31, 2017
Project Status	Completed
Approved Budget	0.00
Project Leader	EDUARDO M. MARIN
Co-Project Leader	N/A
Members	

Date of Preparation	Sep 06, 2019

Project Significance

Project Objectives

Project Outputs & Performance Indicators	
Expected Output (Deliverable)	**Performance Indicators**
Verified modified pretreatment process for pineapple fiber	Verified modified pretreatment process for pineapple fiber
Verified pretreatment process for pineapple fiber using spent liquor	Verified pretreatment process for pineapple fiber using spent liquor
Technical paper for publication/IP application (UM claim process)	Technical paper for publication/IP application (UM claim process)

Year #	Funding Source Name	PS		MOOE		CO	TOTAL
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost		
Total Cost		-	-	-	-	-	-

PS Breakdown						
Budget Item (LIB)		Amount				TOTAL
Staff Name	% Involvement	**Q1**	**Q2**	**Q3**	**Q4**	
Total Cost		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

MOOE Breakdown					
Budget Item (LIB)	Amount				TOTAL
	Q1	**Q2**	**Q3**	**Q4**	
Total Cost	**0**	**0**	**0**	**0**	**0**

Work Plan

Completion by Quarter

#	Major Activity	Milestone Indicator	Date Completion

Risk Management Plan		
Key Success Factor	**Risk**	**Countermeasure**

Change Control Plan

Quality Management Plan

Management & Organization

Manpower

Training & Development